

Pre-Analysis Plan + Identification Memo

NextGen (NGN) Empirical Finance Research Camp · Prof. Lei Gao (George Mason University) Week 7 (Independent Research Project I) — assigned Fri Aug 7, 2026 · DUE before the Week-8 session, Friday, August 14, 2026. Bring it ready to present — you'll defend one line of it out loud.

— Why this assignment

You have a topic, a verified literature map, and a toolkit. This is the assignment where they become a **project**. You will write the two documents that turn a hunch into a design a skeptic has to take seriously: a **Pre-Analysis Plan (PAP)** and an **Identification Memo**. Together they are the **design half of your capstone paper** — the half that is worth more than the results, because a clean design makes any result (including a null) informative, and a dirty design makes even a "significant" one worthless.

The discipline here is the same one the whole camp has been building toward: **commit before you peek**. You will state your hypotheses, your one primary specification, and your falsification tests **before** you run a single outcome regression. The point is not to guess right; it is to make your test honest, so that whatever it returns, you — and your reader — can believe it.

One rule above all others: file the plan **before** the first result. The value of a pre-analysis plan is entirely in its timing.

— Your task

Take **your own project** — the one behind your topic and literature map — and produce three short pieces.

Part 1 — The Identification Memo (the design)

Write, for your design:

(a) The identifying-assumption sentence. One sentence, filled in: > "Comparing [treated units] to [controls], my estimate of [the effect] is causal if [the identifying assumption] holds, which is plausible because [the design feature that makes it plausible]."

(b) The threats-and-responses table. At least **four rows**. For each threat to that assumption, name the threat, say why it endangers the estimate, and give your **design response**.

Threat	Why it endangers the estimate	Design response
Confounding trend	...	...
Reverse causality	...	...
Selection into treatment	...	...
Measurement error	...	...

(c) The identifying variation. One or two sentences: what variation in the data actually identifies your effect? (A policy adopted at different times? A discontinuity? An instrument? A

Part 2 — The Pre-Analysis Plan (Olken short form)

Keep it to **one page**. State, in this order: 1. **Hypotheses** — numbered and **directional** (e.g., "H1: the ban raises municipal yields; $\beta_1 > 0$ "). Use a one-sided test only if you truly held the prior before seeing data. 2. **The primary specification** — exactly one. Name the **outcome**, the **treatment**, the **sample**, the **controls**, the **fixed effects**, the **clustering level**, and the **estimator**. Write it as an equation and in words. 3. **Secondary / heterogeneity tests** — pre-specified splits (and only these count as confirmatory). 4. **Multiple-testing control** — if you test more than one outcome, state your **Benjamini-Hochberg FDR** correction. 5. **Falsification / placebo tests** — at least one test that **should return nothing** if your design is clean (a placebo outcome, a pre-treatment "effect," an unaffected control group). 6. **What would falsify H1** — finish the sentence: "I will conclude my hypothesis is wrong if ..."

Part 3 — The reproducible data-pull plan

A short paragraph or table: the **data source** (WRDS/CRSP/Compustat, EDGAR, FRED, HMDA, 13F, PatentsView, yfinance, ...), the **access route**, the **pinned vintage/date**, the **unit of observation**, the **sample window**, the **key variables**, any **identifier crosswalk** you need (PERMNO/GVKEY/CIK), and how you will make the pull **reproducible** (cache-or-fetch, an audit log, no hand-editing). Name one **known limitation** of the data.

— What to hand in

1. The **Identification Memo** — the assumption sentence + a ≥ 4 -row threats table + the identifying variation. 2. The **Pre-Analysis Plan** — the six items above, on one page. 3. The **data-pull plan** — source, vintage, unit, window, variables, crosswalk, reproducibility, one limitation.

Format: ~1.5–2 pages total (tables are fine). **Due:** before the Week-8 session, **Friday, August 14, 2026**. **Bring it ready to present** — you'll read your identifying-assumption sentence aloud and take one threat from the room.

— A worked example (model answer, abbreviated)

Topic: do state same-sex-marriage bans raise municipal borrowing costs? (Modeled on Gao, Liu & Wang, "The Rainbow of Municipal Credit.")

Identification Memo — assumption sentence. > Comparing states that adopt a same-sex-marriage ban to states that have not-yet adopted one, my estimate of the effect on municipal yields is causal if ban and non-ban states' yields would have trended in parallel absent the ban, which is plausible because adoption timing is driven by court and legislative calendars rather than by a state's credit trajectory.

Threats-and-responses (excerpt).

Threat	Why it endangers the estimate	Design response
Ban states differ in fiscal trend	A pre-existing yield trend would masquerade as a "ban effect"	Event-study pre-trends; not-yet-treated controls; Callaway–Sant'Anna

A contemporaneous state policy	Another policy at the same date confounds the ban	Placebo on an unrelated outcome; controls for known concurrent policies
Staggered timing biases TWFE	"Forbidden comparisons" put negative weights on the estimate	Use a heterogeneity-robust estimator (CS), not vanilla two-way FE

Pre-Analysis Plan (excerpt).

- **H1 (directional):** a ban raises the state's municipal offering yield; $\beta_1 > 0$.
- **Primary spec:** $\text{Yield}_{it} = \beta_1 \cdot \text{Ban}_{it} + \alpha_i + \delta_t + \varepsilon_{it}$, on state-month issuance, **state and month fixed effects**, standard errors **clustered by state**, estimated with **Callaway-Sant'Anna** (not-yet-treated controls).
- **Falsification:** no "effect" should appear in the **pre-adoption** window; a placebo ban date two years early should return null.
- **Falsifies H1:** I will conclude my hypothesis is wrong if the CS estimate is a precisely-estimated zero, or if pre-trends are non-flat.

Notice what the example does not contain: a result. That is the point — every choice above is fixed before the first outcome regression.

— How you'll be graded

Short rubric — you're doing great if:

- Your **identifying-assumption sentence** is one clean sentence with all four blanks filled — and the "plausible because" is a real design feature, not a wish.
- Your **threats table** has ≥ 4 threats, each with a specific design response (not "add controls").
- Your **PAP names exactly one primary specification**, fully specified (outcome, treatment, sample, controls, FE, clustering, estimator) — a reader could run it without asking you a question.
- You include **at least one real falsification test** and finish the "I will conclude I'm wrong if..." sentence.
- Your **data-pull plan** names a source, a pinned vintage, the unit/window, and one honest limitation — and is reproducible.
- The whole thing reads like it was written **before** you saw any results. (It should have been.)

— Fillable template

Project title / question: _____

Part 1 — Identification Memo

- **Assumption sentence:** Comparing _____ to _____, my estimate of _____ is causal **if** _____, which is plausible because _____.
- **Identifying variation:** _____

Threat	Why it endangers the estimate	Design response

Part 2 — Pre-Analysis Plan

- **H1 (directional):** _____ · **H2 (optional):** _____
- **Primary specification (equation):** _____
- **Primary specification (words):** outcome = _____ · treatment = _____ · sample = _____ · controls = _____ · fixed effects = _____ · clustered by = _____ · estimator = _____
- **Pre-specified heterogeneity:** _____
- **Multiple-testing control (if >1 outcome):** BH-FDR at $q =$ _____
- **Falsification / placebo test:** _____
- **I will conclude H1 is wrong if:** _____

Part 3 — Reproducible data-pull plan

- **Source & access:** _____ · **Pinned vintage/date:** _____
- **Unit of observation:** _____ · **Sample window:** _____
- **Key variables:** _____ · **Identifier crosswalk:** _____
- **Reproducibility (cache / audit / no hand-edits):** _____
- **One known limitation:** _____

— Tips

- **Write the assumption sentence first.** If you can't fill in "plausible because ____" with a design feature, you don't yet have identification — you have a correlation with hope attached.
- **One primary spec, not five.** The forking-paths problem is real: pick the specification you'd defend to a referee and pre-commit to it. Everything else is secondary.
- **Falsification is a gift.** A placebo that could have failed and didn't is the strongest thing in your paper. Design one you might actually fail.
- **Pin the data vintage.** "The CFPB Data Browser snapshot as of Aug 2026" — the same query returns different numbers across re-publications; name yours.
- **Match the estimator to the design.** Staggered timing → Callaway-Sant'Anna, not vanilla TWFE. A one-day price reaction → an event study, not a multi-period DiD.
- **This is your paper's spine.** A clean PAP now is your August 15 methods section already half-written.

You've got this — see you Friday, August 14.